

A Sharp Barrier for Size-Only Control of Local Oscillation

Liang Wang
Huazhong University of Science and Technology

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Abstract

The triangle inequality gives only block length after q/T normalization. A constant-sign synthetic sequence attains that bound, so boundedness and support size alone cannot yield any fixed- X power saving. This stops only the size-only local oscillation method. The classification is OBSTRUCTION_L1; no L2 arithmetic progress or endpoint credit is claimed.

1 Frozen target

The selected route is `0161.bad_endpoint_pointwise_fixed_atom` [1]. The required signature is actual fixed- h_0 packet, named fixed atom, deterministic every prefix, deterministic every scale, fixed- X power at that atom, and actual active support. Throughout $h_0 = 2$ is only a source-backed data fact.

2 Exact result

Theorem 1. *For $|a_j| \leq 1$ on a block of length L , $\max_r \sum_j \leq r a_j \leq L$, with equality for $a_j = 1$. Hence no uniform $X^\wedge$ -sigma follows from coefficient size and cardinality alone.*

Proof. Triangle inequality gives $\max_r \leq L \sum_j \leq r a_j \leq L$ whenever $|a_j| \leq 1$. The synthetic fixture $a_j = 1$ attains L at $r = L$. Therefore no argument using only these two premises can force a factor $X^\wedge$ -sigma; arithmetic cancellation is an additional necessary input. $\square$

The smallest missing literal theorem is ARITHMETIC_CANCELLATION_INPUT_FOR_LOCAL_INCREMENTS.

3 Scoped stop and claim firewall

The cell `SIZE_ONLY_LOCAL_OSCILLATION_METHOD` is STOP-SCOPED. This does not stop either `O161` pointwise parent or the global architecture.

The result is L0/L1 only. Phase averages and Lebesgue-almost-every phase are not named atoms. Fixed $h_0 = 2$ is not decay. Archive addressing is not canonical physical representation, and empty-domain truth is not actual active support. The named-atom endpoint charge is zero, so the strict 1/400 budget is unpaid. We claim no program-positive L2, prime-pair lower bound, or twin-prime theorem.

4 Reproducibility

The adjacent JSON payload records the full six-axis signature, source lock, typed result, exact scoped-stop cell, endpoint ledger, and claim firewall. The audit artifact contains eight true mutation-rejection assertions.

References

- [1] Liang Wang. Mvp8 source-phase route decision, 2026.